Review Article

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Research progress on knowledge discovery in traditional Chinese medicine medical records based on artificial intelligence

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Abstract

Traditional Chinese Medicine (TCM) medical records represent the clinical practice records of physicians, encapsulating a wealth of TCM clinical diagnosis and treatment experiences as well as academic thoughts of numerous TCM practitioners. Such various information provides certain basic conditions for the integration of artificial intelligence (AI) technology with TCM. Through analyzing successful clinical records, AI technology can extract practical and understandable knowledge. Therefore, this paper summarizes and analyzes the application of information technology in the field of knowledge discovery research from Chinese medical records. It focuses on 3 key aspects: the resources of TCM Medical Records, the objects of TCM records mining, and the application of AI technology. The aim is to provide beneficial references for maximizing the value of TCM medical records.

Keywords: artificial intelligence, TCM medical records, knowledge discovery, research progress

Introduction

Traditional Chinese medicine (TCM) medical records, often referred to as "Yi An" in Chinese, are individual case studies that encapsulate the clinical experience and academic thoughts of TCM practitioners. Historically, research on these medical records has predominantly relied on personal experience and interpretive analysis methods, which are subject to the subjective influences of the researchers and focus on case-by-case analysis. The process is limited by human resources and time constraints, making it challenging to identify underlying patterns from a large volume of medical records. However, the advancement of information science and technology has provided the means to swiftly unearth the clinical experience of TCM practitioners from vast amounts of medical record data. The structured nature of TCM medical records also facilitates their processing by computers. These records are compiled based on the comprehensive application of TCM theories and therapeutics, ensuring a degree of authenticity and covering a wide range of disease types and treatment methods, thus providing the fundamental conditions for the integration of TCM medical records with information processing technologies.

Artificial Intelligence (AI) is a technological domain dedicated to emulating human sensory capabilities and behavioral patterns through the principles of bionics, neurology, statistics, and other disciplines[1]. Medicine, as one of its application fields[2], has shown its unique value and potential, especially in aiding clinical diagnosis. When combined with the essence and characteristics of TCM medical records, AI can make up for the human brain's functional shortcomings in dealing with massive medical records. Further, it can excavate valuable, practical, and understandable knowledge from a plethora of effective clinical medical records, converting the individual subjective insights of many physicians into collective experiences depicted by objective data[3]. This paper provides a summary and analysis of the application of information processing technologies, including AI, in the field of knowledge discovery from TCM medical records. It focuses on 3 aspects: the resources of TCM medical records, the objects of TCM records mining, and the application of AI technology, aiming to offer useful references for maximizing the value of TCM medical records.

Resources of TCM medical records

TCM records from renowned traditional Chinese medicine practitioners

Distinguished TCM practitioners represent a unique group in the TCM domain, symbolizing the highest standards of TCM scholarship and clinical expertise. Strengthening research on academic ideas and the inheritance of experience from these eminent TCM practitioners can better carry forward and enhance TCM studies, making a contribution to the development of the TCM industry. This area plays a central role in the field of knowledge discovery research from TCM medical records. The research methodology mainly revolves around the exploration and analysis based on the

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Guidelines and Standards in Chinese Medicine, 2024, 2(4):174–181

Received 3 August 2024; Accepted 30 September 2024

Published online 7 October 2024

http://dx.doi.org/10.1097/gscm.0000000000000035

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clinical experiences, medication patterns, and academic philosophies of specific TCM physicians. For example, the study of national TCM masters such as Zhongying Zhou[4], Zhizheng Lu[5], and Lei Zhang[6], employs visualization technology and data mining to analyze their clinical medication practices and academic insights.

TCM records of specific diseases

In the realm of knowledge discovery research derived from TCM medical records, those primarily categorized by specific diseases can be divided into 2 main categories: TCM disease nomenclature and western medical disease nomenclature. Conditions named according to TCM include "Pi Man[7]," "Xiong Bi[8]," and "Zhong Feng[9]," while those named according to western medicine include coronary heart disease[10], threatened miscarriage[11], and neurodermatitis[12]. These medical records mainly investigate medicinal use and therapeutic regulations, such as prevalent symptoms, syndrome patterns, and formulae, in hopes of offering innovative approaches for treating modern diseases with TCM.

In addition, there are medical records that are based on TCM departmental divisions or guided by specific TCM theories, aiming to provide insights into the medication practices of clinicians. Examples include records based on the concurrent treatment of qi and Blood[13], and records of unstable angina pectoris diagnosed under the meridian theory[14], etc., which analyze the patterns of diagnosis and treatment and the experience with medication for specific diseases.

TCM records of specific prescription and TCM herbs types

The distinctiveness of the TCM theoretical framework, particularly its principle of "same treatment for different disease," suggests that identical treatment protocols may be applied to various diseases under identical pathologic conditions. This notion has inspired numerous researchers to concentrate not only on medical records as clues but also to explore medical records categorized by prescription and TCM herbs types, aiming to uncover the universal patterns of TCM therapy and to further enrich and evolve TCM diagnosis and treatment strategies. Researchers excavate medical records of specific prescription types based on classic TCM formulas, analogous prescriptions, specific formulas, or empirical formulas of Chinese medicinal compounds. This method primarily focuses on the relationship between prescriptions and syndromes, embodying the TCM concept of "same treatment for different diseases." For instance, data mining and analysis of related treatment medical records for classic formulas such as "Xiao Chai Hu Tang[15]," "Xiao Qing Long Tang[16]," construction and visual analysis of knowledge graphs for thirst-quenching formulas[17] and classic prescriptions[18]. By examining the clinical application patterns of frequently used classic and specific prescriptions, it offers guidance for clinicians on the appropriate application of prescriptions with corresponding syndromes and the reference for adjusting the use of medicinal ingredients. In addition, the exploration of "symptom-medicine" and "disease-medicine" relationships in some analogous prescription medical records is also conducted, such as those similar to Xiao Chai Hu Decoction[19].

Medical records can also originate from specific TCM herbs, such as records in *Ju Tong Wu's medical records* involving *cassia twig* prescriptions[20], and records in "The essence of modern Chinese eminent physicians' medical records" where *cinnamon* is first used in the initial or follow-up visits[21], to investigate the clinical application characteristics and compatibility patterns of different medicinal substances.

TCM records from specific historical eras and geographic regions

Throughout the extensive history of TCM's evolution, it is evident that as eras have changed, a variety of diagnosis standards and levels of comprehension regarding the same illnesses have emerged among practitioners. This variety is equally evident in the understanding and application of medicinal prescriptions. Each medical expert, drawing from their clinical experiences and theoretical insights, possesses unique perspectives and practices in selecting treatments and medications. Consequently, medical records from various time periods, with the aid of AI, can uncover novel medical perspectives from those specific times. For instance, employing data mining techniques to investigate the characteristics of medications used for chronic kidney disease during the Ming and Qing dynasties[22], and the particularities of prescriptions for spring warmth conditions in Qing dynasty medical records[23].

Furthermore, given China's vast geography and the substantial climatic and environmental variations across its regions, distinct medical schools have arisen, such as the Hejian School, Yishui School, Yongjia School, Qiantang School, Lingnan School, and Xin'an School[24]. By utilizing data mining and other methodologies, one can delve into the experiences of esteemed TCM practitioners in Jiangsu, Zhejiang, and Shanghai with plaster formulas[25], the therapeutic approaches of Xin'an medical practitioners to lung expansion[26], and the distinctive practices of renowned Lingnan TCM practitioners in managing chronic liver diseases[27], etc.

Objects of TCM records mining

The TCM medical records encompass intricate and multifaceted information. To unlock their profound value, it is essential to take diverse viewpoints for a detailed examination and manipulation of the different data sets within these records. Utilizing dimensionality reduction techniques allows for more efficient identification and selection of suitable approaches, thereby defining the goals of knowledge discovery research in medical records.

Disease names

In the context of TCM medical records, the designation of a disease plays a crucial role in the diagnosis process. Reflecting the varying perceptions of illnesses within TCM and Western Medicine frameworks, investigating these disease names can offer theoretical backing and practical direction for the modernization and globalization of TCM. The research in this sector is primarily centered on medical records that share similar etiological factors or prescription types, scrutinizing the prevalent western disease categories under the guidance of TCM theory.

For instance, studies based on the clinical use of shepherd's purse flower by Zhongying Zhou[28] have summarized that the primary diseases targeted by shepherd's purse flower are nephritis, with a focus on treating kidney deficiency and damp-heat. Analysis of modern medical records utilizing the formula of

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Bupleurum and Cinnamon Twig Decoction[29] reveals that the predominant diseases treated are related to the digestive system. Exploration of modern gastroesophageal reflux disease records[30] delves into the primary TCM-related conditions, such as stomach ache, epigastric pain, and stomach distension.

Symptoms

Symptoms, including the synthesis of the 4 diagnosis methods, constitute the basis for syndrome differentiation and treatment in TCM. Under the TCM theoretical framework, symptoms are closely related to syndromes. Symptoms form the basis of syndromes, and syndromes are the direct targets of treatment. Numerous researchers have employed methodologies such as "symptom-herbal" and "syndrome-symptom" to elucidate the specific processes of syndrome differentiation and treatment, as well as the legislative prescription approaches of physicians in medical records. For instance, Jiang and Yan[31] conducted a study on the syndrome-symptom-herbal patterns in medical records associated with "half-exterior and half-interior" identifying a set of potential symptom combinations. This indicates that although Xiao Chai Hu Decoction is the principal formula for treating conditions of half-exterior and half-interior, the scope of the syndrome for this condition is broader than that covered by Xiao Chai Hu Decoction syndrome. Some researchers, leveraging modern clinical compilations from renowned TCM practitioners, have established a knowledge structure system for the terminology of symptoms related to heart diseases in TCM clinical practice[32], thereby laying the groundwork for the standardization and normalization of TCM symptom terminology.

Since tongue diagnosis and pulse diagnosis are integral components of the TCM diagnosis framework, they offer a direct reflection of the state and progression of diseases. Some researchers have dedicatedly investigated the correlation between various tongue and pulse characteristics and medication in clinical medical records. For instance, Wu et al[33] conducted a study on the medical records of phlegm-heat syndrome documented by Mengying Wang, with a particular focus on his tongue records. They delved into his diagnosis approach and treatment rationale for this syndrome, analyzing the patterns evident in his tongue diagnosis, syndrome differentiation, and prescription practices. Wu[34] has analyzed and discussed the association between tongue and pulse signs and medication in the context of infantile diseases across historical medical records. Ye et al[35] have explored the efficacy evaluation methods of TCM for treating chronic obstructive pulmonary disease (COPD). They have worked on standardizing the terminology related to the progression of the disease in relation to its symptoms and severity. Their findings suggest that prominent contemporary TCM practitioners have mainly focused on relieving particular systemic symptoms associated with COPD. This research established a foundation for the study of TCM clinical efficacy assessment methods.

Syndromes and pathomechanisms

Syndromes and pathomechanisms constitute pivotal elements in the modernization of TCM, bridging clinical applications with foundational theories. Pathomechanisms underlie the development of syndromes, while syndromes reflect the clinical manifestation of pathomechanisms. By analyzing syndromes, researchers can deduce the pathomechanisms of illnesses. The present research focuses on the interplay between the holistic syndrome or pathomechanism and the use of medication. Thereby, they can enhance the clinical comprehension of specific etiologies and pathomechanisms. For instance, studies on Professor Zhongying Zhou's application of black soybeans in relevant medical records[36] and the clinical records of contemporary TCM gynecology experts treating dysmenorrhea[37] involve a multifaceted analysis that encompasses diagnosis, pathomechanism, and syndromes, alongside the comprehension of diseases and clinical pharmaceutical applications. This approach aims to deepen the understanding of TCM in relation to specific conditions or medicinal substances.

Incorporating AI and other information processing technologies to summarize the distribution characteristics of disease syndromes, to clarify the classification methods of syndromes, and to investigate the regularities of syndromes can aid in the standardization and quantification of syndrome diagnosis[38]. For instance, based on clinical medical records of TCM syndromes in atrial fibrillation[39], it is concluded that the syndromes of qi-yin deficiency and qi deficiency with blood stasis are the most prevalent. In the record of lung expansion medical records[26], multidimensional research is conducted from symptoms, syndromes, and medications to explore and analyze the associations between lung expansion syndromes and symptoms and core medications. Some researchers, based on ancient medical texts on cough records in TCM[40], have analyzed the distribution patterns of cough syndromes. Employing the method of inferring syndromes from prescriptions, they identified nine common cough syndromes and discovered 2 new syndrome types: external cold with internal dampness and phlegm-stasis obstructing the lung.

It is evident that targeting syndromes for excavation can refine the syndrome typing of certain modern diseases, offer insights for the explicitation of TCM's implicit knowledge, and provide a reference method for exploring the relationships between syndromes and disease symptoms and core medications. It can facilitate the development of valuable new prescriptions. Beyond using syndromes as indicators, some researchers delve into the elements of syndromes to uncover their composition, distribution, and evolutionary patterns, and subsequently investigate the rules of combining these syndrome elements[41]. For example, in the medical records of renowned TCM practitioners treating coronary heart disease[42], the top 3 syndrome elements are blood stasis, qi deficiency, and phlegm turbidity. They also investigate the primary lesion location elements and nature elements in chronic atrophic gastritis and precancerous gastric conditions[43].

Prescriptions and TCM herbs

The prescriptions and medicinal herbs documented in medical records represent the externalization of the therapeutic approaches administered by practitioners based on their clinical experience and through the process of disease diagnosis and syndrome differentiation. Prescriptions are carefully crafted combinations of various herbs, formulated in accordance with TCM theory, where the inclusion, exclusion, and dosage of each herb can significantly impact the clinical outcomes. AI and other information processing technologies excel at managing objective and quantifiable data, meaning that they are particularly well-suited for this task.

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Researchers who primarily target prescriptions as the main subject of study often aim to uncover new formulations from existing medical records or to reinterpret classic and commonly used prescriptions under the guidance of TCM theory. For instance, they delve into the core formulations, drug combinations, and novel prescriptions for treating hypertensive vertigo[44]. By examining medical records documented in modern clinical research literature spanning the past 2 decades, they investigate the primary diseases treated with these classic prescriptions and the main adverse symptoms encountered in contemporary clinical practice[45].

TCM herbs exhibit unique attributes, such as the 4 energies and 5 flavors, which hold particular significance for in-depth exploration. For example, Lv and Xu[46] have investigated the compatibility patterns of medicinals used by Tianshi Ye for treating vomiting, focusing on their properties and flavors. Yan et al[47], based on records of gastric pain, suggest that renowned TCM practitioners skillfully employ medications that counteract each other, not strictly adhering to the rigid constraints of "the Pharmacopoeia of the People's Republic of China." Some researchers have also investigated the link between modern medical indicators and Chinese medicinal prescriptions in contemporary medical records. For example, Zheng et al[48] have used association rule methods to analyze the medication patterns during the stage of spleen damp-heat syndrome, gathering and analyzing patients' laboratory tests and prescriptions from medical records. Their analytical approach merits our close attention and consideration for adoption.

Application technologies of artificial intelligence

At present, the utilization of AI within the domain of TCM includes a range of technologies, such as Natural Language Processing (NLP), Data Mining, Machine Learning, Deep Learning, Knowledge Graphs, and Large Language Models. These technologies are not typically used in isolation but are often integrated to accommodate diverse research requirements and inquiries. Practical applications extend to areas like disease prediction and prevention, clinical decision-making support for diagnosis and treatment, and health management within TCM. The application of AI to research in TCM knowledge discovery, which entails extracting valuable insights and knowledge from extensive data sets such as medical records, is predominantly centered on Data Mining and the construction of Knowledge Graphs.

Machine learning and data mining

Machine learning stands as a subdivision of AI, employing mathematical theories and methodologies to build and train models for data mining, thereby facilitating the learning from and prediction of data[49]. Data mining involves the extraction of implicit, previously unidentified, yet potentially valuable information and knowledge from vast, incomplete, noisy, ambiguous, and stochastic data sets, constituting an essential phase in the realm of knowledge discovery[50].

Frequency analysis method

Frequency analysis is a commonly utilized and simple technique in the research of TCM medical records, providing the most intuitive reflection of medication patterns for specific diseases or the prescribing habits of physicians. For example, Wang[51] and colleagues employed the frequency analysis method, based on ancient medical records related to depressive symptoms, identifying common high-frequency medicinals, the properties, and flavors associated with Chinese medicine, and summarizing their frequently used drug combinations and key prescription components.

Unsupervised learning methods

Unsupervised learning is a branch of machine learning that is mainly applied in situation of lacking target attributes, with the feature that there are no labels or results in the training data sets. It is generally used to analyze the relationships and clustering within data sets, including methods such as association rules, clustering analysis, factor analysis, and principal component analysis[52]. These algorithms discover associations and clusters within the data without the need for any predefined targets, revealing latent knowledge and experiences within the framework of TCM[53]. For example, Zhu and Zhan[54], based on medical records from the Menghe School for treating emotional disorders, used unsupervised entropy-based hierarchical clustering analysis to investigate the medicinal usage patterns of the Menghe practitioners in treating such disorders. Dong and Zhu[55], based on Professor Zhongying Zhou's medical records for treating breast cancer, applied intraset association rule methods to dissect the medical records in terms of clinical symptoms, pathogenesis, and medication use, and to summarize the patterns of syndrome differentiation and treatment.

Supervised learning methods

Beyond the previously mentioned data mining techniques, models can also be constructed by analyzing TCM medical records and subsequently applied in areas such as AI and disease diagnosis and treatment. Supervised learning methods involve the creation of models for classifying unknown data, which primarily include Logistic Regression Analysis, Bayesian Networks, Discriminant Analysis, and Artificial Neural Networks[52]. Supervised learning is mainly used to address classification and regression issues and is appropriate for research on "disease-symptom-syndrome" relationships. It holds greater value in the development of clinical syndrome differentiation models for common diseases, risk prediction models, and research on risk factors for syndrome elements[53]. Chen et al[56], in their exploration of the distribution patterns of cough syndromes in ancient TCM medical records, utilized a combination of hidden structure and Logistic Regression Analysis to uncover several new cough syndromes. Zhao et al[57] employed decision tree and neural network methods to establish a diagnosis model for the deficiency of yin and yang in hypertension patients. Gan et al[58], based on Professor Ziyu Wang's medical records for treating endometriosis, concluded that their results were entirely in line with Professor Wang's academic philosophy regarding the treatment of endometriosis, demonstrating the utility of Bayesian Networks in the transmission of expertise from senior practitioners.

Deep learning and transfer learning are also involved in the data mining methods of TCM medical records. These methodologies are frequently combined to tackle classification and

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regression problems, particularly in the extraction and recognition of TCM tongue images. Liu et al[59] have explored the application effects of deep learning and transfer learning technologies in extracting and identifying local features of TCM teeth marks and tongue fissures. Utilizing these technologies has effectively completed the task of identifying local features of TCM tongue images. The conclusions drawn from medical records research, with the increasing application of various models, can expand the perspectives and application domains of medical records inheritance and development in a more objective and scientific way.

Integration of data mining with other research methods

Present researchers are employing not only data mining techniques to analyze TCM medical records but also trending towards integrating other methodologies for a comprehensive analysis, such as incorporating pharmacological mechanisms from network pharmacology. For example, Wang et al[14] have utilized data mining to analyze and organize modern medical records of unstable angina, investigating the patterns of syndrome differentiation and treatment. They have combined network pharmacology to delve into the molecular mechanisms by which core medicinal pairs in the treatment affect unstable angina. Zeng et al[60] have applied multiple algorithms to uncover medication patterns for diabetes treatment and have explored the pharmacological mechanisms of key formulations in treating diabetes using bioinformatics, offering guidance for the syndrome-based prevention and management of diabetes.

Knowledge graphs

Knowledge graphs are essentially complex semantic networks consisting of numerous nodes and edges. Nodes in these graphs represent concrete entities, while edges indicate either the attributes of these entities or the connections between them[61]. At present, many TCM researchers are creating TCM knowledge graphs by drawing on a variety of sources, such as medical records, ancient texts, and contemporary literature, or by fusing multiple data types. Potential sources of data for TCM medical records knowledge graphs encompass specialized clinical data for certain diseases, records from esteemed TCM practitioners, electronic medical records from specific medical departments, online data, and historical medical records. These graphs are utilized for the development of TCM knowledge graphs, investigation into TCM knowledge discovery, the transmission of expertise from distinguished TCM practitioners and the conversion of their implicit knowledge into explicit forms, the creation of knowledge-based Question Answering systems, and the exploration of decision support avenues.

For instance, Yu et al[62] have preliminarily established a TCM clinical knowledge graph based on clinical knowledge sources such as TCM medical records. This graph is composed of core concepts, including diseases, syndromes, symptoms, prescriptions, and Chinese medicinal herbs. Li[63], using the medical records of 3 TCM practitioners who diagnose and treat insomnia through syndrome differentiation and treatment, has created a corresponding database. He has combined knowledge graph construction techniques with the efficacy evaluation of syndrome differentiation and treatment, resulting in the development of a knowledge graph that integrates complex network optimization strategies. This has laid a foundational groundwork for the construction of TCM records knowledge graphs.

Some researchers have focused on the medical records of a specific physician, employing knowledge graph technology to explore the latent clinical diagnosis and therapeutic experiences within these records. For example, by analyzing the clinical approach and characteristics of researcher Jingqing Hu's treatment of dementia using knowledge graph analysis techniques, a dementia diagnosis and treatment knowledge graph has been constructed, which follows the structure of "staging-syndrome-clinical manifestations-prescriptions-herbs[64]." The medical records from "Xugao Wang's medical records"[65] have been presented using knowledge graph technology to reveal their explicit knowledge, while also analyzing the underlying logical reasoning relationships to deduce Wang's experience in treating consumptive diseases. Knowledge graph and data mining technologies have been combined with the medical records of Professor Zhongying Zhou in treating lung cancer[66] and Zhizheng Lu in treating Sjögren's syndrome[5], to achieve a visual representation of a physician's clinical experience and to explore the relationships between symptoms, pathogenesis, and prescriptions in the medical records.

In addition, researchers have focused on the medical records of a group of physicians, using knowledge graph technology to seek consensus on the experience of treating certain diseases with TCM. For example, the treatment of type 2 diabetes with qi and yin deficiency syndrome[67], vascular dementia[68], renowned TCM practitioners' treatment of coronary heart disease[69], and lung cancer[70], among others.

Deep learning and large language models

Large language models refer to deep learning models with an extensive network scale, characterized by a large number of model parameters (typically in the hundreds of billions). These models are trained on vast amounts of general text data to learn patterns and relationships within language[71]. Medical large language models have garnered widespread attention and application both domestically and internationally for their extensive research and impressive performance in tasks such as medical information extraction, question-answering systems, and knowledge graph construction. They can serve as knowledge-based auxiliary tools for physicians, demonstrating significant potential in medical practice, research, and education[72].

Currently, the application of large language models in TCM involves areas such as intelligent diagnosis and treatment, knowledge question-answering, prescription generation, and medicine recommendation[73]. For instance, the Zhongjing TCM large language model[74], the Qi Huang Dao Asks large model, and the Huangdi model[75] are all in the initial stages of exploration. The training data for these models come from various medical knowledge literature, and the existing applications in medical records text include entity extraction and terminology standardization[76], providing support for further research on medical records.

Summary and prospects

From the perspective of the resources of medical records, most current research is based on the records of renowned TCM

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practitioners. This approach allows for a comprehensive analysis of a particular physician's overall diagnosis and treatment thought process, providing a unique avenue for discovering the experiences of these eminent TCM practitioners. When classified by the nature of the medical records themselves, disease-based classification is currently the most widespread method. It can fully reflect the specific diagnosis, treatment thought process, and medication patterns of a particular disease. However, its drawback is that it does not reflect the clinical thinking of TCM's syndrome differentiation and treatment, as well as the principle of treating different diseases with similar syndromes. Classifying medical records based on prescriptions and medicines has the advantage of reflecting the characteristics of TCM diagnosis and treatment from the perspective of treating different diseases with similar syndromes. To some extent, it can compensate for the shortcomings of disease-based classification, but it cannot fully reflect the clinical experience of the physician. Classifying medical records based on region and period gives them unique regional and temporal characteristics. However, due to the difficult-to-understand classic Chinese language of ancient medical records, as well as the vast geographical areas and numerous dialects, researchers face significant challenges in data preprocessing.

From the perspective of medical records mining, this aspect is essentially about how researchers deconstruct and analyze medical records based on their research objectives. Most existing researchers primarily focus on uncovering and analyzing the overall clinical treatment experience of physicians for specific diseases, providing new practical ideas for clinical practitioners. However, there is a scarcity of research aimed at the reverse validation of existing TCM theories using modern computer information processing technologies. In the context of modern medical fields, such as western medicine diseases and indicators, the research forms are often singular and limited, necessitating further exploration.

From the perspective of applied AI, current information processing methods are diverse, and each has its own advantages. The discovery of TCM experiential knowledge in medical records can be focused on methods such as data mining and knowledge graph construction. Researchers should apply specific methods based on the specific circumstances encountered during their research process to leverage their strengths. Data mining is a broad concept that encompasses a multitude of techniques and methods, focusing on discovering potential patterns and regularities from a large volume of medical records. Knowledge graphs organize the information in medical records in the form of graphs, emphasizing the visualization and analytical reasoning of the knowledge contained in the records. They can also be used in combination in the research of medical records knowledge discovery, complementing each other.

Currently, the methodologies and technologies still follow the thinking patterns of Western medicine, which makes it difficult to meet the complex and multidimensional relationships of disease, symptoms, syndromes, and medication within the TCM system. This approach can weaken the diagnosis and treatment ideas guided by TCM theory. Some researchers combine expert interviews and other methods to validate the results obtained, avoiding the pitfall of relying solely on computer analysis. The reliability of the content output by large language models largely depends on the quality of the training data, which makes its application in medical knowledge discovery relatively limited. In the pursuit of discovering correct clinical diagnosis and treatment knowledge from medical records, professional domain knowledge graphs, and other knowledge base technologies can provide specialized background information and reasoning basis for large models. Effectively combining the different technological advantages in the field of AI can more efficiently promote the progress of knowledge discovery research in TCM medical records within the domain of AI.

This article summarizes the current state of knowledge discovery research combining artificial intelligence technology with TCM medical records from the aforementioned 3 aspects. AI has made significant contributions to the modernization of TCM. Existing technologies, such as deep learning and large language models, as part of the emerging AI, are still in the initial exploratory stage of research and application in the field of TCM and have great prospects for development and application. Combining the rich data sources in the TCM field, such as medical records, it is believed that in the future, AI technology will provide more significant progress and breakthroughs in the inheritance and exploration of TCM knowledge.

Acknowledgments

Thank you to all contributing authors, with special thanks to Z.Y. and T.X. for their guidance on this article.

Statement of ethics

None.

Conflict of interest disclosure

The authors declare no competing interests.

Funding source

This work was supported by the Central Level Public Welfare Scientific Research Institution Basic Scientific Research Business Expense Personal Project (YZX-202313): Methodological Research on the Discovery of Traditional Chinese Medicine Theoretical Knowledge Based on Clinical Medical Cases; Central Level Public Welfare Scientific Research Institution Basic Scientific Research Business Expense Key Support Project (YZX-202330): Research on the Construction of Title Page a Knowledge Graph of Insomnia Syndrome in Traditional Chinese Medical Cases.

Data availability statement

Data sharing is not applicable to this article as no data sets were generated or analyzed during the current study.

Author contributions

Z.T. and Q.W.: contributed equally to this work. Y.J., S.D., and G.F.: gave proposals. J.Y.: assisted with the writing. X.T. and Y. Z.: revised the manuscript.

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How to cite this article: Tang Z, Wu Q, Jia Y, et al. Research progress on knowledge discovery in traditional Chinese medicine medical records based on artificial intelligence. *Guidelines Stand Chin Med.* 2024;2(4):174–181. http://dx.doi.org/10.1097/gscm.0000000000000035

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